

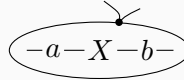
Round 2: every approach targets the book's *current* notation

All approaches now share the book's real `tikz-styles.tex` (`triangle`, `copydot`, `\dloop`, `\dwhiskers`). Test: eq. (70)-style derivative row with the Penrose loop.

A. Legs as node options

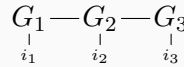
A1: derivative of $a^T X b$ (Penrose loop)

```
\begin{tikzpicture}[tn]
  \node[legs={180}] (a) {$a$};
  \node[right=.7em of a] (X) {$X$};
  \node[right=.7em of X, legs={0}] (b) {
    $b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(a)(b)}
  \dwhiskers
\end{tikzpicture}
```



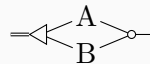
A2: TT chain

```
\begin{tikzpicture}[tn]
  \node[legs={270/i_1}] (G1) {$G_1$};
  \node[right=1em of G1, legs={270/i_2}] (
    G2) {$G_2$};
  \node[right=1em of G2, legs={270/i_3}] (
    G3) {$G_3$};
  \draw (G1) -- (G2) -- (G3);
\end{tikzpicture}
```



A3: Khatri–Rao $A * B$ (book geometry)

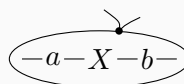
```
\begin{tikzpicture}[tn]
  \node[triangle, rotate=180] (f) {};
  \node (A) at (.6,.25) {A};
  \node (B) at (.6,-.25) {B};
  \node[copydot, legs={0}] (c) at (1.2,0)
    {};
  \draw (f) -- (A) -- (c);
  \draw (f) -- (B) -- (c);
  \draw[double] (f) -- ++(-.4,0);
\end{tikzpicture}
```



B. The graphs library

B1: chain + loop drawn after

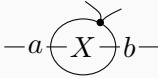
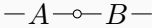
```
\begin{tikzpicture}[tn]
\graph[grow right sep=.7em] {
  l[as=, coordinate] -- a[as=$a$] --
  X[as=$X$] -- b[as=$b$] --
  r[as=, coordinate]
};
\dloop[55]{(a)(b)}
\dwhiskers
\end{tikzpicture}
```



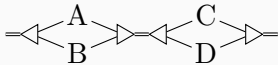
B3: Khatri–Rao fan — does not work. Two attempts (`branch down sep`, `explicit yshifts`) could not

make the `graphs` placement engine center a fan-out/fan-in between two triangles; the branch nodes land on or below the chain axis and downstream nodes misalign. Verdict: approach B is pleasant for *chains* (B1 composes fine with `\dloop`), but branching shapes should use approach A or D.

C. Mini-DSL (scope: inline chains)

C1: derivative row — loop via trailing ’	
<code>\tn{-a-{X’}-b-}</code>	
C2: Hadamard-style chain	
<code>\tn{-A-*-B-}</code>	

D. Pics (molecules)

D2: mixed product (511) by composition	
<pre> \begin{tikzpicture}[tn] \pic (k1) {kron={A}{B}}; \pic (k2) at (1.7,0) {kron={C}{D}}; \draw[double] (k1-in) -- ++(-.35,0); \draw[double] (k1-out) -- (k2-in); \draw[double] (k2-out) -- ++(.35,0); \end{tikzpicture} </pre>	
D3: determinant stack	
<pre> \begin{tikzpicture}[tn] \pic {det stack=X}; \end{tikzpicture} </pre>	

E. Paper reproduction test (approach A + `\dloop`)

The forms the book actually uses, including subset loops.

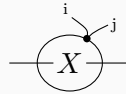
R1: eq (70) full derivation — whole loop, X-only loop, result

```
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=.7em of a] (X) {$X$};
  \node[right=.7em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(a)(b)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=1em of a] (X) {$X$};
  \node[right=1em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
$=$
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=2.2em of a] (b) {$b$};
  \draw (a.east) -- ++(.25,.28);
  \draw (b.west) -- ++(-.25,.28);
\end{tikzpicture}
```

$$a-X-b = a \text{---} X \text{---} b = a \text{---} \text{---} b$$

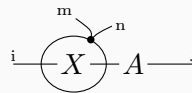
R2: eq (73) — lone X, labeled whiskers, free edges through the loop

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \draw (X) -- ++(-.8,0);
  \draw (X) -- ++(.8,0);
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
    ++(-.2,.28)
    node[pos=1, anchor=south east, font=\tiny] {i};
  \draw (ddot) .. controls +(45:.12) ..
    ++(.28,.18)
    node[pos=1, anchor=west, font=\tiny] {j};
\end{tikzpicture}
```



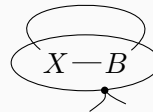
R3: eq (74) — X-only loop inside a longer chain

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (A) {$A$};
  \draw (X) -- ++(-.8,0)
    node[pos=1, above, font=\tiny] {i};
  \draw (X) -- (A);
  \draw (A) -- ++(.8,0)
    node[pos=1, above, font=\tiny] {j};
  \dloop[55]{(X)}
  \draw (ddot) .. controls +(80:.12) ..
    ++(-.2,.28)
    node[pos=1, anchor=south east, font=\tiny] {m};
  \draw (ddot) .. controls +(45:.12) ..
    ++(.28,.18)
    node[pos=1, anchor=west, font=\tiny] {n};
\end{tikzpicture}
```



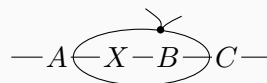
R4: trace row (108) — arc crossing the loop, dot below

```
\begin{tikzpicture}[tn]
  \node (X) {$X$};
  \node[right=1em of X] (B) {$B$};
  \draw (X) -- (B);
  \path (X) edge[out=140, in=40, looseness=2.6] (B);
  \dloop[-55]{(X)(B)}
  \dwhiskersdown
\end{tikzpicture}
```



R5: subset loop spanning two nodes mid-chain

```
\begin{tikzpicture}[tn]
  \node (A) {$A$};
  \node[right=1em of A] (X) {$X$};
  \node[right=.7em of X] (B) {$B$};
  \node[right=1em of B] (C) {$C$};
  \draw (A) -- (X) -- (B) -- (C);
  \draw (A) -- ++(-.6,0);
  \draw (C) -- ++(.6,0);
  \dloop[55]{(X)(B)}
  \dwhiskers
\end{tikzpicture}
```



R6: splitting-rule mid-step — loop around an inverse

```
\begin{tikzpicture}[tn]
  \node (a) {$a$};
  \node[right=1em of a] (X) {$X^{-1}$};
  \node[right=1em of X] (b) {$b$};
  \draw (a) -- (X) -- (b);
  \dloop[55]{(X)}
  \dwhiskers
\end{tikzpicture}
```

